Ajay Venkatesh

Brooklyn, NY

+1 516 585 9013

Education

New York University | *Master of Science in Computer Engineering (GPA – 3.9/4)* Aug 2024 – May 2026

Skills

Programming Languages: Python, Java, C++, JavaScript (ES6), TypeScript, SQL
AI/ML: PyTorch, LLM Fine-tuning, RAG (LangChain, FAISS), NLP, Hugging Face, SVM, Random Forest, MLflow, PySpark
Web Development: REST API, GraphQL API, Microservices, React, Node.js, Express.js, Spring Boot
Database: RDBMS (MySQL, PostgreSQL), NoSQL (MongoDB, JSON), Dataverse
Cloud: AWS (AWS CDK, API Gateway, Lambda, IAM, Q, SageMaker), Azure (AI services, Function Apps, Storage blobs)
Tools: Git, Jupyter, Postman, Docker, GitLab CI, Terraform, Ansible, Boto3, Datadog, Power Automate, Power BI, Agentic AI Workflows, MCP Servers

Experience

Campus Mesh **Jan 2026 – Present**
AI Software Engineer *New York, US*

- Developed and maintained scalable **Node.js microservices** powering real-time academic workflows for **10K+ users**, including study groups, messaging, notifications, and profile management.
- Designed and implemented a collaborative-filtering **recommendation engine** using user interaction signals (course views, clicks, and engagement data), increasing platform engagement by 28% and user retention by 50%.
- Authored **LLM-based retrieval pipelines** for CampusIQ (AI Assistant), improving contextual accuracy by **30%** through semantic retrieval and structured data indexing.

Amazon **May 2025 – Aug 2025**
Software Development Engineer Intern *Seattle, US*

- Built a high-throughput, event-driven incident triage system using AWS ECS Fargate Tasks, SQS, and SNS, decoupling message processing and reducing incident response latency by **50%** during peak traffic.
- Deployed a **Retrieval-Augmented Generation (RAG)** pipeline with an agentic workflow that autonomously retrieves operational logs to reason over incident details, generating context-aware summaries and remediation steps.
- Improved large-scale log analysis by implementing a **Model Context Protocol (MCP)** server for efficient retrieval, achieving a **40%** reduction in AWS Athena query time and compute costs.
- Provisioned resilient **Infrastructure-as-Code (IaC)** using AWS CDK to manage auto-scaling ECS clusters, ensuring 99.9% availability under 2x projected load.

New York University (Novel AI Technologies, Inc.) **Aug 2024 – Present**
AI Software Engineer *New York, US*

- Architected an **NSF-funded ($100K)** AI-powered health platform for **iOS** and **Android** ingesting real-time wearable telemetry; implemented **time-series anomaly detection** using **Isolation Forest and LSTM models**.
- Built **computer vision pipelines (YOLOv8, EfficientNet)** to process user-submitted food images integrated with **GPT-4 workflows** for nutritional estimation and dietary feedback.
- Developed an insurance analytics platform **(React, Node.js, REST APIs)** integrating **LLM-based document analysis (GPT-4 + LangChain)** for underwriting insights.
- Engineered ETL pipelines transforming **NoSQL** data into **PostgreSQL** with concurrency control; processed **250K+** records and powered **AWS QuickSight** dashboards.

PricewaterhouseCoopers (PwC) **Aug 2021 – Aug 2024**
Senior Software Engineer *Bangalore, India*

- Designed and implemented a **distributed backend system on Microsoft Azure** integrating with **Microsoft Dynamics** and supporting ML inference pipelines for healthcare analytics.
- Led **code reviews** and mentored interns on software design and **test-driven development**, reducing production bugs by **25%**.
- Built an **NLP-powered OCR pipeline** using **LayoutLM and Azure Form Recognizer** to automate data extraction from medical documents, reducing processing time by **75%**.
- Optimized data ingestion using **parallel processing** for **100K+ Excel records**, reducing execution time from **5 hours to 1 hour**.
- Developed **Power BI dashboards** integrating multiple enterprise databases, improving reporting efficiency by **40%**.

Projects

InviLite (AI-Driven Data Engineering & Analytics Platform) | *AWS, SageMaker, Bedrock, FAISS, Python*

- Architected a cloud-native data infrastructure (Kinesis, Lambda, S3 Bronze/Silver/Gold, Glue, DynamoDB) ingesting structured and unstructured enterprise data into governed ML pipelines.
- Built a RAG system (SVM classification + FAISS vector indexing + Bedrock LLM) enabling contextual resolution recommendations.
- Reduced incident resolution time from 1–2 hours to minutes and improved historical match accuracy by 30%.
- Increased root-cause classification accuracy from 56% to 92% through structured feature engineering and model retraining.
- Implemented dual-layer (real-time + batch) Lambda architecture supporting continuous learning and inference scalability.

Certifications & Publications

AWS Cloud Practitioner
Power BI Data Analyst Associate
Stanford Machine Learning (Coursera)
Azure Data Fundamentals

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques *(IEEE)*
An Aquila-optimized SVM classifier for Diabetes prediction *(IEEE)*